BRD — Customer Retention Analytics Engine

Business Requirements Document

1. Executive summary

Customer attrition reduces monthly recurring revenue (MRR). This initiative uses the Telco Customer Churn dataset and a Random Forest classifier to identify customers predicted to churn, quantify the associated monthly-revenue exposure, and enable focused retention action through a Power BI dashboard.

The dashboard is a decision-support product: it must distinguish observed historical churn from a model prediction and must not claim that a prediction is a cancellation.

2. Business objectives

- Identify customers with a high predicted likelihood of churn for prioritised retention outreach.
- Quantify monthly revenue at risk from the predicted at-risk segment.
- Enable segmentation by the exported service, contract, payment and demographic proxy fields.
- Target a 10% reduction in overall churn during Q3, subject to a defined campaign baseline, ownership, and measurement design.

3. Stakeholders

Stakeholder	Responsibility
Executive sponsor / Commercial lead	Sets retention target and approves campaign investment
Marketing / Retention	Designs offers, prioritises lists, records contact outcomes
Customer success / Sales	Executes outreach and captures disposition
Business analyst / BI intern	Maintains requirements, dashboard and data validation
Data analyst / ML owner	Re-runs model, monitors quality, documents performance

Finance	Validates MRR-at-risk and realised retained revenue
Data governance / Privacy	Approves access, retention and customer-data handling

4. Scope

In scope: historical churn modelling; export of test-set model results; a Power BI decision dashboard; customer-segment analysis; monthly revenue-at-risk calculation; documented campaign workflow.

Out of scope: real-time scoring, automated customer communications, causal attribution of retention offers, customer lifetime-value modelling, and a promise that an individual prediction will occur within 30 days. The supplied code trains on historical labels and produces a holdout prediction; it does not contain an event-time churn horizon.

5. Business requirements

ID	Requirement	Acceptance criterion
BR-01	Provide a curated model-output CSV for Power BI.	CSV opens without an index column and includes `Actual_Churn`, `Predicted_Churn`, `MonthlyCharges`, plus exported feature fields.
BR-02	Surface at-risk volume and monthly MRR exposure.	KPI cards reconcile to the imported data under current filters.
BR-03	Support action-oriented segmentation.	Users can compare predicted churn by encoded contract, internet service and other available export columns.
BR-04	Preserve prediction versus actual-label clarity.	All visual titles and tooltips name the metric explicitly; no “confirmed churn” label is used for predictions.

BR-05	Support campaign prioritisation.	A table can filter to `Predicted_Churn = 1` and export a worklist if a customer identifier is available.
BR-06	Enable performance review.	Confusion-matrix counts, precision, recall and accuracy can be calculated from the export where both target and prediction exist.
BR-07	Make ROI assumptions transparent.	Dashboard exposes a user-controlled retention save-rate assumption and shows potential retained MRR, not realised savings.

6. KPI definitions

KPI	Definition
Evaluated customers	Count of rows in the exported holdout dataset
Predicted at-risk customers	Count where `Predicted_Churn = 1`
Predicted risk rate	Predicted at-risk customers / evaluated customers
Monthly revenue at risk	Sum of `MonthlyCharges` where `Predicted_Churn = 1`
Actual churn rate	Count where `Actual_Churn = 1` / evaluated customers
True positives / false positives / false negatives / true negatives	Counts from `Actual_Churn` and `Predicted_Churn`
Precision	True positives / predicted at-risk customers
Recall	True positives / actual churners

Potential retained MRR	Monthly revenue at risk × approved retention save-rate assumption

Do not publish numeric model performance or ROI claims until refreshed from the actual exported CSV and reconciled to the model run.

7. Business process and decisions

1. Load approved CRM-history extract.
2. Validate completeness and clean charge values.
3. Score the holdout population with the approved model.
4. Publish the model-output CSV and refresh Power BI.
5. Review predicted at-risk segments and MRR exposure.
6. Select an approved campaign audience; exclude consent-restricted or ineligible customers.
7. Execute retention outreach and log offer/contact outcomes.
8. Compare subsequent actual churn and retained revenue against a baseline/control design where available.

8. Success criteria

- Refresh completes with no schema/type errors.
- KPI totals reconcile to CSV-level calculations.
- Stakeholders can identify highest-exposure segments and create a governed action list.
- Campaign reporting separates potential retained MRR from realised retention results.
- Model performance is documented for the scoring run before operational use.

9. Risks and controls

Risk	Control
Encoded columns are hard for business users to read	Rename them in Power Query; preserve source names in a data dictionary.
Export omits customer ID	Use segment-level analysis only; retain ID in a governed non-model feature export if individual outreach is required.
Predictions treated as facts	Use “predicted” terminology and show precision/recall.

Revenue at risk overstated as savings	Label it exposure; calculate potential benefit with a transparent save-rate parameter.
Model drifts	Re-score on an agreed cadence and monitor performance once newer labels arrive.